

FROM ZERO TO TRADING BOTS

Workshop PINTU x Web3 Dev Bandung

Sabtu, 25 Juli 2026 | 13:00 WIB | Parla Cafe, Bandung

Instructor: Putra

"Ga perlu jadi programmer. Cukup prompt, download, cron, jalan."

1. Persiapan — Install Dari Rumah

Penting: WiFi venue mungkin lambat. Install ini dari rumah biar ga buang waktu saat workshop berlangsung.

1.1 Python 3.10+

Download: <https://www.python.org/downloads/>

Cek setelah install (harus Python 3.10.x atau lebih baru):

```
python --version
```

PENTING: Centang "Add Python to PATH" pas install!

1.2 curl

- **Windows 10/11:** Sudah include. Cek dengan `curl --version`
- **Mac:** Sudah pre-install.
- **Linux:** `sudo apt install curl`

1.3 Code Editor

Pilih salah satu sesuai kenyamanan:

- **VS Code:** <https://code.visualstudio.com/>
- **Cursor:** <https://cursor.sh/>
- **Notepad++:** Kalau ingin yang ringan saja.

1.4 Setup Provider — OpenRouter (Gratis)

Kenapa OpenRouter? Gratis, tidak perlu isi saldo, dan mendukung banyak model (DeepSeek, Qwen, Gemini, Llama).

1. Buka <https://openrouter.ai/keys>
2. Klik "Sign Up" — login menggunakan Google / GitHub / email.
3. Setelah login, klik "Create Key".
4. Beri nama key: **Workshop PINTU** (bebas).
5. Centang semua permission, lalu klik "Create".
6. **Copy API key** yang muncul dan simpan di Notepad.

Alternatif (kalau gagal daftar OpenRouter):

- **Gemini API:** <https://aistudio.google.com/apikey> (gratis 60 req/menit)
- **Ollama (local):** install + download model gratis (butuh RAM 8GB+)

1.5 Install Hermes Agent

Buka terminal (CMD / PowerShell / Terminal Mac/Linux):

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

Lalu cek instalasi:

```
hermes doctor
```

Catatan Windows: Butuh PowerShell sebagai Administrator kalau ada error permission.

1.6 Akun Telegram

Download Telegram di HP. Bikin username. Nanti ini akan dipakai untuk menerima notifikasi bot.

2. Panduan Workshop Lengkap

2.1 Kenapa Bot Sinyal?

Lo ga perlu jadi programmer buat bikin signal bot. Cukup bisa buka terminal, bisa copy-paste, dan mau nyobain. AI assistant (Hermes Agent) yang nanganin sisanya: *Download template → jalanin → cron-in → sinyal masuk HP.*

Hasil akhirnya: Tiap X jam, bot analisa pasar, generate sinyal, kirim ke Telegram. Lo tinggal baca, eksekusi kalau cocok.

Penting: Ini signal bot, bukan execution bot. Bot kita kasih sinyal beli/jual — lo sendiri yang eksekusi di exchange. Aman, ga konek API exchange.

2.2 Kenapa Hermes Agent?

- **Open source:** gratis, ga ada biaya langganan.
- **Cron bawaan:** bot jalan otomatis tanpa server eksternal.
- **Multi-provider:** bisa pake Claude, GPT, DeepSeek, bebas.
- **Multi-platform:** output ke Telegram, Discord, file, atau email.

2.3 Setup Provider di Hermes

```
hermes model
```

Pilih: **OpenRouter**. Paste API key yang udah di-copy tadi. Pilih model: **DeepSeek V3** (gratis, cepat, cukup buat workshop).

Model	Bagus Buat
DeepSeek V3	Signal analisa, prompt umum
Qwen 2.5 72B	Coding sederhana
Gemini Flash	Gratis unlimited

2.4 Jalanin Bot Template

Download `trading_bot_template.py` dari link workshop. Jalanin:

```
python trading_bot_template.py
```

Output yang diharapkan:

```
TRADING SIGNAL 25 Jul 13:30 WIB
Price: $4,075.70 (+1.71%)
Direction: LONG
Confidence: 95%
Entry: $4,075.70
SL: $3,962.50 | TP: $4,113.20
R:R: 1:1.67
Reasons:
- RSI > 50 bullish bias
- EMA9 > 21 bullish
- MACD+ bullish momentum
```

Kalau bot silent (ga ada output)? Itu normal. Bot punya cooldown 4 jam ga akan generate sinyal kalau masih ada posisi aktif atau baru aja generate. Cek history sinyal: `cat signals.json`

2.5 Cron — Otomatisasi

Manual python tiap hari? Capek. Cron = bot jalan sendiri, lo tinggal terima hasil.

```
hermes cron create 'every 4h' --name "Sinyal Auto" --prompt "Jalankan python trading_bot_template.py 2>&1. Kirim output apa adanya."
```

Cek cron: `hermes cron list`

2.6 Kirim ke Telegram

```
hermes gateway setup
```

Pilih Telegram, bikin bot lewat **@BotFather** di Telegram, dapetin token. Start bot di HP. Update cron biar deliver ke Telegram:

```
hermes cron update <job_id> --deliver telegram
```

Tiap 4 jam, sinyal masuk HP. Lo ga perlu buka laptop.

3. Trading Bot Template (Full Code)

File: `trading_bot_template.py`

```
#!/usr/bin/env python3
# FROM ZERO TO TRADING BOTS - Workshop PINTU x Web3 Dev Bandung
# Bot sederhana: ambil harga -> hitung indikator -> generate sinyal.

import json, math, urllib.request
from datetime import datetime, timezone, timedelta

# Config
COOLDOWN_HRS = 4
MAX_HIST = 100
INDICATORS = {
    "RSI": "Relative Strength Index (momentum)",
    "EMA": "Exponential Moving Average (trend)",
    "MACD": "Moving Average Convergence Divergence",
    "BB": "Bollinger Bands (volatility)",
    "ATR": "Average True Range (SL/TP base)"
}

# 1. Price Fetch
def fetch_price():
    # TODO: Ganti dengan API real (Kitco, Binance API, CCXT)
    # Demo data:
    return {
        "bid": 4075.70,
        "ask": 4076.10,
        "change_pct": 1.71,
        "timestamp": datetime.now(timezone.utc).isoformat()
    }

# 2. Helpers
def load_json(path, default):
    try:
        with open(path) as f: return json.load(f)
    except (FileNotFoundError, json.JSONDecodeError):
        return default

def save_json(path, data):
    with open(path, "w") as f:
        json.dump(data, f, indent=2)

# 3. Indicators
def calc_rsi(prices, period=14):
    if len(prices) < period + 1: return 50.0
    gains = losses = 0.0
    for i in range(-period, 0):
        d = prices[i] - prices[i-1]
```

```

        if d >= 0: gains += d
        else: losses -= d
    ag, al = gains/period, losses/period
    if al == 0: return 100.0
    rs = ag / al
    return 100.0 - (100.0 / (1.0 + rs))

def calc_ema(prices, period):
    if len(prices) < period: return prices[-1] if prices else 0
    k = 2.0 / (period + 1.0)
    ema = sum(prices[-period:]) / period
    for p in prices[-period:]:
        ema = p * k + ema * (1.0 - k)
    return ema

def calc_macd(prices, fast=12, slow=26, signal=9):
    if len(prices) < slow + signal: return 0.0
    return calc_ema(prices, fast) - calc_ema(prices, slow)

def calc_bollinger(prices, period=20, std_dev=2.0):
    recent = prices[-period:]
    sma = sum(recent) / period
    var = sum((p - sma)**2 for p in recent) / period
    std = math.sqrt(var)
    return sma, sma + (std_dev * std), sma - (std_dev * std)

def calc_atr(history, period=14):
    if len(history) < period: return 10.0
    ranges = []
    for i in range(-period, 0):
        h = history[i].get("high", history[i].get("bid"))
        l = history[i].get("low", history[i].get("bid"))
        ranges.append(h - l)
    return sum(ranges) / period if ranges else 10.0

# 4. Generate Signal Core Logic
def generate_signal(prices, history, last_signal):
    # Cek posisi & cooldown
    if last_signal and last_signal.get("status") == "active": return None
    if last_signal:
        last_time = datetime.fromisoformat(last_signal["timestamp"])
        elapsed = (datetime.now(timezone.utc) - last_time).total_seconds()/3600
        if elapsed < COOLDOWN_HRS: return None

    price = prices[-1] if isinstance(prices[-1], (int, float)) else
prices[-1].get("bid")

    rsi = calc_rsi(prices)
    ema9, ema21 = calc_ema(prices, 9), calc_ema(prices, 21)
    macd = calc_macd(prices)
    bb_mid, bb_up, bb_lo = calc_bollinger(prices)
    atr = calc_atr(history)

```

```

reasons = []
conf = 50 # Base confidence

# RSI Logic
if rsi >= 70: reasons.append(f"RSI ({rsi:.0f}) >= 70 overbought"); conf -= 20
elif rsi <= 30: reasons.append(f"RSI ({rsi:.0f}) <= 30 oversold"); conf += 20
elif rsi > 50: reasons.append(f"RSI ({rsi:.0f}) > 50 bullish"); conf += 5
else: reasons.append(f"RSI ({rsi:.0f}) < 50 bearish"); conf -= 2

# EMA & MACD Logic
if ema9 > ema21: reasons.append("EMA9 > 21 bullish"); conf += 12
else: reasons.append("EMA9 < 21 bearish"); conf -= 12

if macd > 0: reasons.append("MACD+ bullish momentum"); conf += 8
else: reasons.append("MACD- bearish momentum"); conf -= 8

# Bollinger Logic
if price <= bb_lo: reasons.append(f"BB lower bounce"); conf += 18
elif price >= bb_up: reasons.append(f"BB upper reject"); conf -= 18

conf = max(0, min(100, conf))
if conf < 50: return None # Terlalu rendah

direction = "LONG" if conf > 50 else "SHORT"
sl_dist, tp_dist = atr * 1.5, atr * 2.5

return {
    "timestamp": datetime.now(timezone.utc).isoformat(),
    "direction": direction,
    "entry": round(price, 2),
    "sl": round(price - sl_dist if direction == "LONG" else price + sl_dist, 2),
    "tp": round(price + tp_dist if direction == "LONG" else price - tp_dist, 2),
    "rr": round(tp_dist/sl_dist, 2),
    "confidence": round(conf),
    "reasons": reasons,
    "status": "active"
}

def format_signal(sig, price):
    lines = [
        "=====",
        f"TRADING SIGNAL | {sig['direction']}",
        f"Entry: ${sig['entry']}",
        f"SL: ${sig['sl']} | TP: ${sig['tp']}",
        f"Confidence: {sig['confidence']}%",
        "====="
    ]
    return "
".join(lines)

def main():

```

```

price = fetch_price()
if not price: return

# Load data
history = load_json("history.json", [])
history.append(price)
history = history[-MAX_HIST:]
save_json("history.json", history)

prices = [h.get("bid", h) for h in history]
signals = load_json("signals.json", [])

# Generate
last_sig = signals[-1] if signals else None
sig = generate_signal(prices, history, last_sig)

if sig:
    signals.append(sig)
    save_json("signals.json", signals)
    print(format_signal(sig, price))

if __name__ == "__main__":
    main()

```

4. Cheatsheet Prompt — Modifikasi Bot

Template bot udah siap. Tapi kalau mau modifikasi, tanya Hermes Agent pake prompt di bawah. Tinggal copas, ganti sesuai kebutuhan, AI akan ubah kodenya.

Contoh Prompt Hermes/AI:

- **Ganti aset (XAU/BTC):** "Ganti fetch_price di trading_bot_template.py supaya ambil harga BTC/USD dari CoinGecko API. Pakai url `https://api.coingecko.com/api/v3/simple/price?ids=bitcoin&vs_currencies=usd`. Parse response JSON untuk dapetin harga."
- **Ubah interval sinyal (cooldown):** "Ubah COOLDOWN_HRS di trading_bot_template.py jadi 2 jam bukan 4 jam."
- **Tambah indikator baru:** "Tambah function calc_stochastic(prices, period=14) ke trading_bot_template.py. Output signal overbought >80, oversold <20."
- **Edit SL/TP rasio:** "Ubah perhitungan SL di trading_bot_template.py jadi 2x ATR, TP jadi 3x ATR."
- **Ganti bahasa:** "Ubah format_signal() di trading_bot_template.py ganti semua tulisan ke Bahasa Indonesia."

Error Umum & Solusi

Error	Solusi
<code>ModuleNotFoundError: No module named 'requests'</code>	<code>pip install requests</code>
<code>SyntaxError</code>	Ada typo di baris kode, paste ke AI minta perbaiki.
Bot jalan tapi ga ada output	Normal. Cooldown masih active / silent mode (belum ada sinyal kuat).

5. Alur Cepat — Ringkasan

1. Install Hermes Agent:

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

2. Setup provider: `hermes model` (pilih OpenRouter → paste API key → DeepSeek V3)

3. Download template: Dapatkan `trading_bot_template.py`

4. Jalanin bot: `python trading_bot_template.py`

5. Cron otomatis: `hermes cron create 'every 6h' --name "Sinyal Saya" --prompt "Jalankan python trading_bot_template.py 2>&1"`

6. Telegram (opsional): `hermes gateway setup`

6. Tips & Best Practices

- **Mulai 1 aset:** XAU atau BTC, jangan semua sekaligus.
- **SL wajib:** Jangan entry tanpa Stop Loss.
- **Backtest:** Pantau performa seminggu sebelum menggunakan real trade (uang asli).
- **Cross-check:** AI bisa salah, cek chart manual.
- **Konsisten:** 3 sinyal bagus > 20 sinyal random. Jangan over-trading (cron tiap jam = spam).
- **Jangan percaya 100% ke AI:** Lo tetap bosnya yang ambil keputusan.

7. Resources

Resource	Link
Hermes Agent	https://hermes-agent.nousresearch.com
OpenRouter	https://openrouter.ai/keys
Python Download	https://www.python.org/downloads/
VS Code	https://code.visualstudio.com/
Telegram Bot	https://t.me/botfather
CoinGecko API (harga)	https://www.coingecko.com/en/api